

commodity taxes. Income tax on individuals and indirect taxes like GST, sales tax, excise duties, etc. on consumer goods. The government impose taxes on the firms as well. Corporation tax is the most important tax falling in this category.

The government spends a part of its tax revenue in purchasing factor services from households and making payments to them in form of wages and salaries.

The government incurs expenditure also in by buying a whole range of goods and services from the firms and makes payments in return.

The government undertakes various social security and welfare schemes in every economy. Old age pension, unemployment allowance, scholarships, etc. are examples of transfer payments received mainly by the households. Similarly government gives subsidies and makes transfer payments to firms in order to encourage production.

Taxes (T) constitute leakages from the flows because they reduce the income and therefore expenditure of the households and firms. The government expenditure (G) is the injection in the flows because it adds to the aggregate demand in the form of government purchases. The equilibrium condition of the circular flow of income in three sector model is:

$$S + T = I + G$$

However, if $(S + T)$ exceed $(I + G)$, income flow decreases because whatever income leaks out from the circular flow of income is not offset by an equivalent amount of injection. If $(I + G)$ exceeds $(S + T)$, the income flow increases because whatever income leaks out from the circular flow of income in the form of $(S + T)$ is less than offset by an injection arising from $(I + G)$.

Operating Surplus: Rent, Royalties, Interest, Profits (dividends, corporation tax + undistributed profits)

Note: The interest in O.S. will not include interest paid on national debt (used for consumption purposes but not production purposes)

MIXED INCOME: Mixed income is earned by self employed.

EXPENDITURE METHOD

$$NNP_{\text{fe}} = \overset{\text{GDP}_{\text{MP}}}{\text{Private final consumption expenditure}} + \overset{\text{(household, club, NGO)}}{\text{govt final consumption expenditure}} + \text{gross domestic capital formation} + \text{Net exports} - \underset{\text{GDP}_{\text{MP}}}{NIT} - \text{Depreciation} + NFIFA$$

Note: • Pvt. final consumption expenditure includes all by households and non-profit institutions on consumer goods.
It excludes all residential investment

- Govt. final consumption expenditure includes current expenditure on goods and services (consumption expenditure) and capital expenditure.
- ~~Govt.~~
- Gross domestic Capital formation includes ~~invest~~ expenditure on residential investment, ^{→ firms} inventory investment (change in stock) and gross fixed business investment.
- Net exports = Exports - Imports

Gross domestic Capital formation = Gross fixed capital formation + change in stock + investment in residential property

$$MU_n = TU_n - TU_{n-1}$$

MU_n is marginally utility of n th unit.

TU_n denotes total utility of ' n ' units.

TU_{n-1} denotes total utility of $n-1$ units.

The relationship b/w Total utility and marginal utility can be indicated in terms of three propositions.

→ The total utility curve is concave from above, indicating declining slope of the curve which means declining marginal utility. Up to the point M, the TV curve has a positive slope, but its slope goes on decreasing steadily as quantity consumed is increased. But marginal utility decreases because total utility increases at decreasing rate.

→ When total utility is maximum, marginal utility is zero. When total utility curve reaches its maximum (at point M), its slope is zero between M and N.

→ After N, the total utility curve has negative slope, showing that the marginal utility is negative.

2. Explain the law of diminishing Marginal Utility.

→ As more and more quantity of a commodity is consumed, the intensity of our desire decreases and, therefore, the utility derived from the additional unit decreases.

→ If there are many uses of a commodity, the most urgent requirement will be fulfilled first, followed by the next important use, and so on.

Theory of Income and Employment

Output / Employment / Income

Aggregate Demand

$$AD = \text{consumption} + \text{Investment} + \text{Govt} + (X - M)$$

closed Economy.

$$AD = C + I$$

AD concept is Ex-Ante (willing).

Ex Ante

planning to buy
demand of the
whole economy

Ex Post

Actually buys

$$= \int (\text{National income})$$

Components of AD.

① Consumption: It refers to the total exp. incurred by the households on the purchase of goods and services in an economy in an accounting year.

Consumption exp. depends on disposable income

② Investment: It refers to the total exp. incurred by all Pvt. firms on capital goods

Eg: machinery, raw materials

Aggregate Supply

AS refers to the money value of goods and services produced by the normal residents of the country in a year.

$$AS = \text{National Income}$$

$$Y = C + S \rightarrow \text{Savings}$$

MC=MR Approach:

Q. When do we decide to produce?

A firm decides to produce when the losses incurred by the firm during production is less than the losses incurred by the firm during shut down.

Cost of production
 $AVC + AFC$

Cost of shut down
 AFC

(i) When $price > AVC$.

loss during production is only a part of AFC .
we can cover full AVC and a part of AFC .

~~Full~~ loss during shut down.
Full AFC .

so the firm will produce

(ii) When $price = AVC$.

loss in production price charged can cover the full AVC ;

Full AFC .

therefore the loss in production is Full AFC .

The loss in production = loss in shut down.

The firm is indifferent but it is judicious to produce.

(iii) When $price < AVC$

loss in production

Full AFC

The price is only able to cover only a part of AVC

∴ The loss in production is a part of AVC and full AFC

∴ The firm will shut down as losses in shut down is less than the losses in production.